


8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- a. We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | x_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | x_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.
- b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | x_i, MARRIED = 0) = \sigma_{SINGLE}^2$ and $\text{var}(e_i | x_i, MARRIED = 1) = \sigma_{MARRIED}^2$. Specify the null hypothesis $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus the alternative hypothesis $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$. We add $FEMALE$ to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE_i + e_i \quad (XR8.6b)$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

- c. Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.
- d. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

a.
$$\begin{cases} H_0: \sigma_M^2 = \sigma_F^2 \\ H_1: \sigma_M^2 \neq \sigma_F^2 \end{cases} \quad \begin{aligned} n_M &= 577 \\ n_F &= 423 \\ SSE_M &= 97161.9174 \\ \hat{\sigma}_F &= 12.024 \end{aligned}$$

$$F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{SSE_M / (n_M - 4)}{(12.024)^2} = \frac{97161.9174 / 573}{144.5766} = 1.1729$$

$$F_{0.975}(573, 419) \approx 1.1968. \\ F_{0.025}(573, 419) \approx 0.8377. \Rightarrow RR: \{F < 0.8377 \text{ or } F > 1.1968\}$$

$$\therefore F = 1.1729 \notin RR, \text{ not reject}$$

b.
$$\begin{cases} H_0: \sigma_{single}^2 = \sigma_{married}^2 \\ H_1: \sigma_{single}^2 \neq \sigma_{married}^2 \end{cases} \quad \begin{aligned} n_S &= 400 \\ n_M &= 600 \\ SSE_{single} &= 56231.0382 \\ SSE_{married} &= 100703.0471 \end{aligned}$$

$$\hat{\sigma}_s^2 = \frac{SSR_s}{df_s} = \frac{56231.0382}{395} \approx 142.357$$

$$\hat{\sigma}_M^2 = \frac{SSE_M}{df_M} = \frac{100703.0471}{595} \approx 169.249$$

$$F^* = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_s^2} = \frac{169.249}{142.357} \approx 1.189 \sim F_{595, 395}$$

$$\text{since } F_{595, 395, 0.025} \approx 0.8338, F_{595, 395, 0.975} \approx 1.1954$$

$$\Rightarrow RR: \{F < 0.8338, F > 1.1954\}$$

$$\because F^* \notin RR \Rightarrow \text{not reject} \#$$

$$c. NR^2 = 59.03 \sim \chi_4^2, \alpha = 0.05$$

$$\chi_{4, 0.95}^2 \approx 9.488 \Rightarrow RR = \{NR^2 > 9.488\}$$

$$\Rightarrow NR^2 = 59.03 > 9.488 \Rightarrow \text{reject, i.e. exist Heteroskedasticity.}$$

In (b), since there is no married in regression, so we can't explain #

$$d. NR^2 = 78.82, df = 4 + 4 + (1/2) = 14 \Rightarrow NR^2 \sim \chi_{14}^2$$

$$\chi_{14, 0.95}^2 \approx 23.685 \Rightarrow RR = \{NR^2 > 23.685\}$$

$$\Rightarrow NR^2 = 78.82 > 23.685 \Rightarrow \text{reject, i.e. exist Heteroskedasticity.}$$

e. The OLS fitted model from part (b), with usual and robust standard errors, is

$$\widehat{WAGE} = -17.77 + 2.50EDUC + 0.23EXPER + 3.23METRO - 4.20FEMALE$$

(se)	(2.36)	(0.14)	(0.031)	(1.05)	(0.81)
(robse)	(2.50)	(0.16)	(0.029)	(0.84)	(0.80)

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

- f. If we add *MARRIED* to the model in part (b), we find that its *t*-value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

e. $\begin{cases} \text{narrower} : \text{Exper, Metro, Female} \\ \text{wider} : \text{EDUC} \end{cases}$

No, there is no inconsistency #

f. The results are completely compatible

∵ (f) shows the marital status doesn't shift the average level of a person's wage.

(b) shows the marital status change the spread or dispersion of a person's wage

∴ it is entirely possible #